

History, Diagnosis, and Management of Chronic Inflammatory Demyelinating Polyradiculoneuropathy



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CME Activity

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Learning Objectives: On completion of this article, you should be able to (1) understand common clinical and electrophysiological features of chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) and pitfalls in its diagnosis, (2) describe typical pathological findings of the nerve in CIDP, and (3) describe typical first-line treatments of

CIDP and options for long-term treatment (including corticosteroid-sparing agents).

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Abstract

Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is probably the best recognized progressive immune-mediated peripheral neuropathy. It is characterized by a symmetrical, motor-predominant peripheral neuropathy that produces both distal and proximal weakness. Large-fiber abnormalities (weakness and ataxia) predominate, whereas small-fiber abnormalities (autonomic and pain) are less common. The pathophysiology of CIDP is inflammatory demyelination that manifests as slowed conduction velocities, temporal dispersion, and conduction block on nerve conduction studies and as segmental demyelination, onion-bulb formation, and endoneurial inflammatory infiltrates on nerve biopsies. Although spinal fluid protein levels are generally elevated, this finding is not specific for the diagnosis of CIDP. Other neuropathies can resemble CIDP, and it is important to identify these to ensure correct treatment of these various conditions. Consequently, metastatic bone surveys (for osteosclerotic myeloma), serum electrophoresis with immunofixation (for monoclonal gammopathies), and human immunodeficiency virus testing should be considered for testing in patients with suspected CIDP. Chronic inflammatory demyelinating polyradiculoneuropathy can present as various subtypes, the most common being the classical symmetrical polyradiculoneuropathy and the next most common being a localized

asymmetrical form, multifocal CIDP. There are 3 well-established, first-line treatments of CIDP—corticosteroids, plasma exchange, and intravenous immunoglobulin—with most experts using intravenous immunoglobulin as first-line therapy. Newer immune-modulating drugs can be used in refractory cases. Treatment response in CIDP should be judged by objective measures (improvement in the neurological or electrophysiological examination), and treatment needs to be individualized to each patient.

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Chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) is an acquired peripheral neuropathy due to an autoimmune attack of peripheral nerve myelin. Because myelin is the main target of the condition, nerve fibers with the most myelin (largest fibers) are the most involved and patients present with weakness, numbness, and sensory ataxia (symptoms of large myelinated fiber dysfunction). The course of CIDP can be varied, and presentations include relapsing-remitting, stepwise progressive, or gradually progressive. The clinical pattern of CIDP is unlike typical peripheral neuropathies that are length-dependent (meaning that the most distal segments are most involved). By contrast, CIDP usually presents as a polyradiculoneuropathy with weakness in both proximal and distal segments (patients have both foot drop and difficulty getting out of chairs). Unlike another immune-mediated demyelinating neuropathy that presents as a monophasic illness with spontaneous recovery, acute inflammatory demyelinating polyradiculoneuropathy (AIDP; also known as Guillain-Barré syndrome), CIDP is a progressive neuropathy that worsens over time. Chronic inflammatory demyelinating polyradiculoneuropathy is responsive to immune-modulating therapy, and controlled trials have shown that corticosteroids, plasmapheresis, and intravenous immunoglobulin (IVIG) are all effective treatments. Chronic inflammatory demyelinating polyradiculoneuropathy usually presents in the “classical” form as a symmetrical disorder with proximal and distal weakness but also can manifest as a variety of other subtypes that can be multifocal or selectively involve sensory or motor nerve fibers. Chronic inflammatory demyelinating polyradiculoneuropathy should be separated from other forms of demyelinating neuropathy: AIDP; monoclonal gammopathy of undetermined significance (MGUS)—associated neuropathies;

human immunodeficiency virus—associated neuropathy; some forms of diabetic neuropathy; uremic neuropathy; inherited peripheral neuropathies (most especially type 1 Charcot-Marie-Tooth disease); and POEMS syndrome (polyneuropathy, organomegaly, endocrinopathy, M-protein, skin changes), a paraneoplastic neuropathy associated with osteosclerotic myeloma or Castleman disease, which is often misdiagnosed as CIDP.

HISTORY OF CIDP

Acquired hypertrophic neuropathies have been recognized for many years, and Austin¹ described cases of probable CIDP in 1958 by recognizing a fluctuating motor-predominant neuropathy that produced severe weakness that would either improve spontaneously or in response to corticosteroids. He noted that some of the cases presented with weakness without muscle atrophy and hypothesized that focal areas of segmental demyelination rather than axonal degeneration were likely the pathological cause because of the lack of atrophy. In a 1975 historical study of 53 personally evaluated patients, Dyck et al² introduced the name chronic inflammatory polyradiculoneuropathy (to which the term *demyelinating* was subsequently added) and thus defined CIDP as a separate disease entity. In this article, the authors described the cardinal clinical, laboratory, electrophysiological, and pathological features of CIDP that are still recognized today. They found that it was a motor-predominant polyradiculoneuropathy that produced proximal and distal weakness and ataxic gait. They also found that there were elevated cerebrospinal fluid (CSF) protein levels and nonuniform slowing of conduction in proximal nerve segments with motor conduction blocks on electrophysiological testing. Pathologically, they found areas of segmental demyelination, onion-bulb formation (stacks of Schwann cell cytoplasmic processes), and mononuclear cell infiltrates, often perivascularly in the

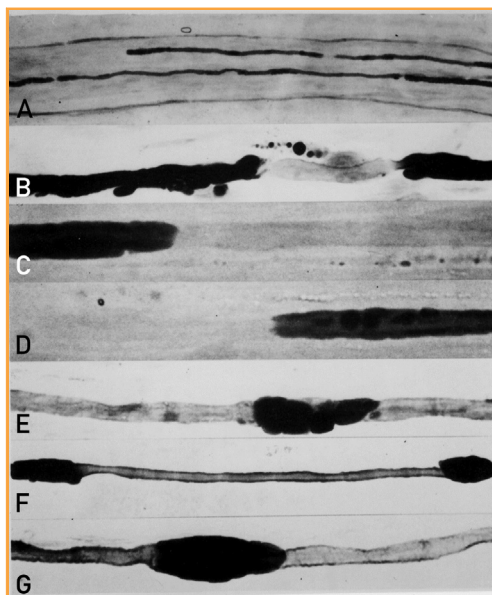


FIGURE 1. Teased fiber abnormalities from the sural nerve of a patient with chronic inflammatory demyelinating polyradiculoneuropathy. A, Small group of myelinated fibers with segmental demyelination and remyelination. B, Demyelinated segment that may already have a few lamellae of new myelin. C and D, Demyelinated segment. E-G, Remyelinated segment with adjacent clumps of degenerating myelin.³

endoneurium and epineurium (Figures 1 and 2).^{2,3} They recognized that CIDP was different from AIDP in that it was progressive and ongoing, and the authors argued that cases with different clinical courses (relapsing-remitting, stepwise progressive, and gradually progressive) were all part of the same disease process because of the shared common features of inflammatory demyelination on pathological and electrophysiological testing as well as elevated CSF protein levels.

CLINICAL FEATURES OF CIDP

In general, CIDP will present with numbness and paresthesia followed shortly by weakness, and these symptoms gradually worsen with time. Patients are weak both distally (eg, intrinsic hand weakness and foot drop) and proximally (eg, difficulty going up stairs or lifting objects onto shelves). In most cases, motor deficits are the most problematic symptoms that patients experience. Patients may have

difficulty walking, often fall, and may require gait aids (eg, wheelchairs); they may also have difficulty with fine finger control. Although decreased sensation and sensory ataxia (not knowing where one's limbs are in space) are common features of CIDP, autonomic symptoms and neuropathic pain are less common. When pain is present in CIDP, it is generally a minor feature (although rare cases can have severe pain). This unequal involvement of different nerve fiber populations (large fiber predominant) likely results from the fact that CIDP is a demyelinating neuropathy, and so the nerve fibers with the most myelin (motor and proprioceptive fibers) are predominantly affected, whereas nerve fibers with little or no myelin (autonomic and pain fibers) are largely unaffected. There is a wide spectrum in the degree in which patients can be afflicted, from minimal involvement (patients are still ambulatory) to severe involvement (patients are wheelchair-bound and unable to feed themselves or attend to their toiletry needs). Respiratory muscles are rarely affected, and it is unusual for patients to be ventilator-dependent. Cranial nerve involvement is also rare, but when it happens, it occurs most commonly in oculomotor and facial nerves.

The justification of the separation of CIDP from AIDP was made mostly because of the different temporal profiles and long-term outcome of these 2 entities. In AIDP, the symptoms and neurological deficits develop quickly over days to weeks and maximal deficits are reached within 1 month. Then gradual improvement occurs and patients get stronger, although long-term residual deficits can persist. By contrast, in CIDP the symptoms and neurological deficits usually evolve slowly over months (at least >2 months). A rapid relapsing-remitting course can also occur in CIDP, and this illness resembles AIDP, but it is different because it is recurrent and not self-limited. This relapsing-remitting course of CIDP is more typical in children and young adults, and it can be separated from AIDP only after patients have multiple attacks.⁴ Ongoing persistent neurological deficits favor CIDP over AIDP, although residual deficits can occur in AIDP. Worsening of the neurological examination over time also favors CIDP. In general, AIDP is a postinfectious neuritis and a preceding infection can often be identified,

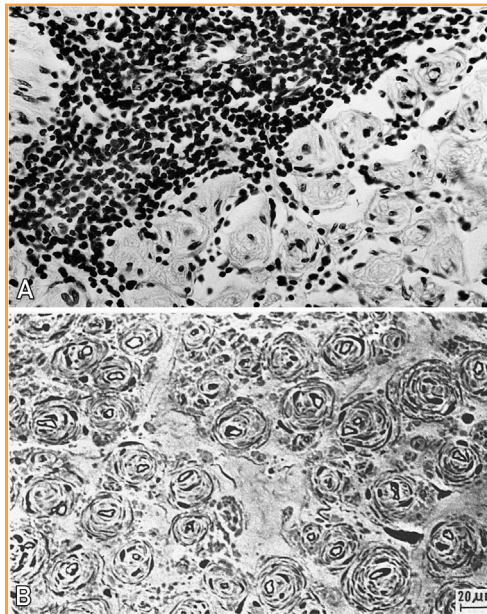


FIGURE 2. Nerve biopsy of chronic inflammatory demyelinating polyradiculoneuropathy. A, Large cluster of endoneurial mononuclear cells associated with a marked degree of onion-bulb formation from the proximal sciatic nerve (paraffin section, hematoxylin and eosin stain). B, Transverse section of a sural nerve to illustrate a marked degree of hypertrophic (onion-bulb) neuropathy and edema (epoxy section, phase contrast).²

whereas in CIDP a preceding infection usually cannot be identified. Part of the inability to recognize a preceding infection in CIDP may relate to the gradual onset of CIDP symptoms that usually cannot be precisely dated.

On neurological examination, motor findings predominate with weakness of shoulder and hip girdle muscles as well as hands and legs. The findings tend to be symmetrical and involve lower and upper limbs. Although the pathological involvement is most severe in proximal nerve segments (roots and plexus), clinical deficits are often worst distally. This is likely because the nerves supplying distal muscles are the longest and so have the most potential exposure to damage by inflammatory demyelination. On sensory examination, proprioception and vibratory detection (large myelinated fibers) are the most severely affected. Deep tendon reflexes are usually absent or severely reduced.

NERVE PATHOLOGY

The pathological damage of CIDP typically affects motor fibers more than sensory fibers, large fibers more than small fibers, and proximal fibers more than distal fibers (mixed motor and sensory nerve roots, plexus and proximal nerves more severely than distal mixed nerves and cutaneous [sensory] nerves). Because of the unequal, patchy nerve involvement, the deficits summate and the longest fibers are the most severely involved.

The typical myelinated nerve fiber changes are segmental demyelination and remyelination that are best seen on teased nerve fiber preparations (Figures 1 and 3).³ Demyelination typically occurs paranodally (near nodes of Ranvier). Nerve fibers remain demyelinated (naked) for only a short time, and Schwann cells quickly remyelinate the formerly demyelinated segment with shorter internodes (intercalated internodes) that have thinner myelin than does the original fiber. Interstitial abnormalities of CIDP include endoneurial edema and inflammatory cell infiltrates (Figures 2 and 3).² Inflammatory infiltrates are typically both endoneurial and epineurial and are frequently perivascular. Demyelination is macrophage-mediated, but finding macrophages actively causing segmental demyelination on nerve biopsy is rarely observed. With repeated demyelination and remyelination, stacks of Schwann cell cytoplasmic processes pile up and form onion bulbs (evidence of repeated long-term demyelination with abortive remyelination) (Figure 3). The involvement of demyelination and remyelination throughout nerves is patchy, with unequal involvement along the length of the nerve. This patchy demyelination involving some fibers more severely than other fibers over distances is the probable explanation for the temporal dispersion observed in nerve conduction studies of CIDP.

Although nerve biopsy can be considered for patients as part of the work-up for CIDP, it is unnecessary for most cases, especially those with typical clinical, electrodiagnostic, and laboratory features of CIDP. In addition, given that demyelination in CIDP is often proximal predominant, it is possible that a distal sensory nerve biopsy (eg, a sural nerve biopsy) will exhibit minimal changes and be nondiagnostic. Biopsy of a more proximal

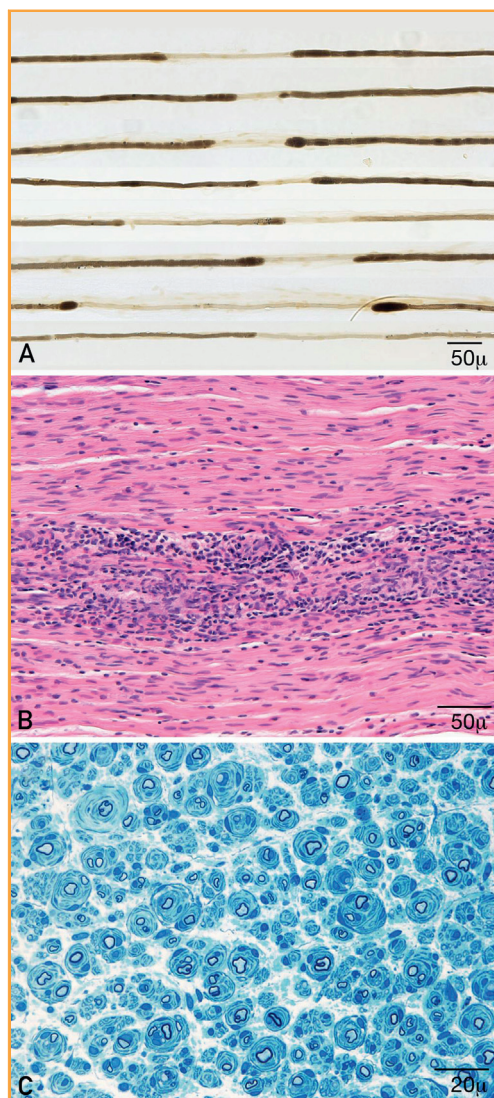


FIGURE 3. Nerve biopsies of patients with chronic inflammatory demyelinating polyradiculoneuropathy. A, Teased fiber preparations from a sural nerve showing multiple segments of segmental demyelination. B, Longitudinal paraffin section (stained with hematoxylin and eosin) from a sciatic biopsy showing large endoneurial perivascular mononuclear cell infiltration. C, Epoxy cross-section (stained with methylene blue) from the sural nerve showing widespread hypertrophic neuropathy with onion-bulb formation (stacks of Schwann cell cytoplasmic processes) and thin myelin. Taken together, these findings are typical of inflammatory demyelination as found in nerve biopsies of chronic inflammatory demyelinating polyradiculoneuropathy.

nerve (eg, at the proximal nerve, plexus, or root level) is generally not necessary unless there is a high suspicion of an alternative diagnosis (especially malignant neoplasm), which would change management because of the increased risk of neurological morbidity from such a procedure. (Proximal biopsies tend to be taken from mixed motor and sensory nerves and may cause new weakness.) Nerve biopsies in possible CIDP cases are generally considered when there is a high suspicion of an infiltrative process such as seen in sarcoidosis, amyloidosis, lymphoma, or another tumor in addition to clues on spinal fluid examination or nerve imaging to suggest the possibility of other disease processes. If nerve biopsy is performed, teased fiber analysis can be helpful, as this can reveal an increased rate of myelin remodeling—segmental demyelination, remyelination, or reduplication of myelin (tomaculae), all of which can be seen in CIDP. Nerve biopsy may also exhibit onion-bulb formation and inflammatory infiltrates. Because of the possibility of sampling error and because of the proximal predilection of many of these pathological abnormalities, the lack of characteristic biopsy features should not dissuade the treating physician from a time-limited treatment trial of immunotherapy if the clinical suspicion remains high.

ELECTROPHYSIOLOGY

All patients suspected of having CIDP should have nerve conduction studies/electromyography performed in addition to a careful history and physical examination. In most cases of typical CIDP, these studies should find evidence of acquired demyelination. Needle electromyography may exhibit only reduced recruitment of motor unit potentials, but depending on the degree of axonal loss, there may also be varying degrees of long duration, high amplitude, complex motor unit potentials, and fibrillation potentials (evidence of active denervation). Nerve conduction studies of CIDP suggest unequal multifocal demyelination in contrast to uniform demyelinating findings typical of inherited neuropathy (eg, type 1 Charcot-Marie-Tooth disease). Features that suggest demyelination include slow conduction

velocities, prolonged distal and peak latencies, prolonged F-wave latencies, and prolonged trigeminal blink responses, conduction block, and temporal dispersion. Frequently upper limb sensory responses are more affected than lower limb sensory (sural) responses, and this observation has been termed “sural nerve sparing” in CIDP. Multiple electrodiagnostic criteria have been developed to diagnose CIDP, with the American Academy of Neurology criteria often used.⁵ More recently, the European Federation of Neurological Societies/Peripheral Nerve Society (EFNS/PNS) clinical electrodiagnostic criteria for diagnosing CIDP were established⁶ and required finding evidence of demyelination of at least 2 nerves for a definite diagnosis of CIDP. Studying more nerves with nerve conduction studies allows for a higher likelihood that the electrophysiological criteria for CIDP can be met. Occasionally, a clinical diagnosis of CIDP will be made when an electrophysiological one cannot be made. The presence of nonuniform features of demyelination in different nerves and the presence of temporal dispersion and/or conduction block are features suggestive of CIDP (acquired demyelinating neuropathy) as opposed to an inherited demyelinating neuropathy.⁷⁻⁹ A notable exception is in connexin-32 inherited neuropathies, which may often have nonuniform demyelinating features.¹⁰ In cases in which demyelination is only proximal (for instance, in chronic immune sensory polyradiculopathy [CISP]; see section on Subtypes of CIDP), somatosensory evoked potentials may be helpful to isolate conduction slowing at the nerve root level. Paying careful attention to nerve conduction abnormalities is important, and 1 group identified 4 major electrophysiological patterns that caused neurologists to misdiagnose other neuropathies as CIDP: (1) a length-dependent axonal neuropathy with mildly slowed conduction velocities being overinterpreted as demyelination, (2) deep peroneal nerve findings from a small foot muscle (extensor digitorum brevis) being interpreted as focal demyelination, (3) mild conduction velocity slowing in motor neuron disease being interpreted as demyelination, and (4) conduction velocity slowing or conduction block at sites of common nerve compression being interpreted as inflammatory demyelination.¹¹

One acquired demyelinating peripheral neuropathy that highly resembles CIDP but has a different pathological basis and requires different treatment is POEMS syndrome. Mauermann et al¹² found that the electrophysiology of POEMS neuropathy has more uniform demyelination, less conduction block, less temporal dispersion, less sural sparing, and more axonal loss than does the electrophysiology of CIDP. Nerve biopsies in POEMS syndrome also have a different pathology than do nerve biopsies in CIDP with less endoneurial inflammation, more axonal degeneration, no onion-bulb formation, and more formation of small epineurial blood vessels (neovascularization).¹³ Another important distinguishing feature is that CIDP should respond to typical immunotherapy (corticosteroids, IVIG, and plasma exchange [PLEX]) whereas POEMS neuropathy does not in a sustained way. If a neuropathy strongly resembles CIDP with demyelinating features electrophysiologically and does not favorably respond to immunotherapy, POEMS syndrome should strongly be considered and an evaluation for the presence of a lambda monoclonal protein, plasmacytoma (with a skeletal bone survey), and titers of vascular endothelial growth factor should be promptly initiated.

EPIDEMIOLOGY AND ROLE OF DIABETES MELLITUS

Estimates of the prevalence of CIDP have ranged from 1.9 cases per 100,000 persons in New South Wales, Australia,¹⁴ to 3.6 cases per 100,000 persons in Italy¹⁵ to 7.7 cases per 100,000 persons in Norway¹⁶ to 8.9 per 100,000 in Rochester, Minnesota.¹⁷ Although historically most experts have thought AIDP to be more common than CIDP, the incidence of CIDP was found to be the same (1.6 persons per 100,000 persons per year) as that of AIDP in the Rochester, Minnesota, population. Some authors have argued that diabetes mellitus is a risk factor for the development of CIDP. Mild demyelinating changes have long been recognized in typical length-dependent diabetic polyneuropathy, with mildly slowed conduction velocities and mildly elevated CSF protein levels also common. So even though the overall pathological process of

diabetic polyneuropathy is more axonal degeneration than demyelination, it is a logical question to ask whether some of these cases could represent CIDP. An epidemiological study performed in the Rochester, Minnesota, population to answer this question found no increased rate of CIDP in patients with diabetes.¹⁷ Similarly, in an Italian cohort, CIDP was not found at an increased frequency in patients with diabetes.¹⁶ A clinical and pathological study of painless motor-predominant diabetic neuropathy questioned whether the cause of these patients' painless motor neuropathy really might be from diabetic CIDP. The pathological findings suggested that it was an axonal and not a demyelinating neuropathy and was caused by ischemic injury and microvasculitis and not by inflammatory demyelinating (as would be expected in CIDP). The authors concluded that painless motor diabetic neuropathy is part of the spectrum of diabetic lumbosacral radiculoplexus neuropathy and does not represent diabetic CIDP.¹⁸ Nonetheless, this is a controversial issue, and many experts still believe that diabetes mellitus is associated with CIDP.

SUBTYPES OF CIDP

Several different subtypes of CIDP exist. The most common presentation of CIDP is the "classical" symmetrical polyradiculoneuropathy described above. The second most common subtype of CIDP is multifocal CIDP, which has also been called Lewis-Sumner syndrome or multifocal acquired demyelinating sensory and motor neuropathy.¹⁹ Lewis et al²⁰ originally described this syndrome as an upper limb—predominant asymmetrical form of CIDP with evidence of focal motor and sensory nerve conduction blocks. Early in the disease course, single nerves are involved, usually beginning with numbness and pain, and over time weakness and atrophy develop. The CSF protein levels tend to be somewhat elevated (not as much as in classical CIDP), and these cases usually respond to corticosteroids (unlike multifocal motor neuropathy [MMN]) and to IVIG. GM1 ganglioside antibodies are not associated with the disease (unlike MMN that are associated). Multifocal CIDP has the same pathology as does classical CIDP with segmental demyelination, epineurial and

endoneurial inflammation, and onion-bulb formation. By contrast, MMN is a pure motor upper limb—predominant neuropathy that has a different underlying pathology than does CIDP (multifocal or classical) and so is likely not a form of CIDP. Taylor et al²¹ did fascicular nerve biopsies at the sites of conduction block in 7 MMN cases and failed to find inflammatory demyelination or onion-bulb formation in any of them. In mild cases, little pathological alterations were seen, and in more severe cases with muscle atrophy, multifocal myelinated fiber loss and regenerating clusters were found.

Another subtype of CIDP is one that is clinically and pathologically confined to the sensory nerve roots and so defies easy diagnosis because the results of nerve conduction studies are normal. Sinnreich et al²² described these cases as CISP and noted that they present with profound sensory ataxia, numbness, and no weakness. The syndrome is one of sensory ataxia (often requiring wheelchairs), large-fiber sensory loss, and frequent falls. Pain is mild. The results of nerve conduction studies/electromyography are completely normal (as the pathology is proximal to the dorsal root ganglion), which have led some of these patients to be falsely labeled as hysterical. Somatosensory evoked potentials can be helpful in exhibiting proximal conduction slowing at the root level. Cerebrospinal fluid protein levels are elevated, and magnetic resonance imaging (MRI) of the lumbar spine exhibits enlarged dorsal nerve roots. Lumbar rootlet biopsies, although not part of the typical work-up when other findings are consistent with this diagnosis, exhibit loss of large myelinated fibers²² (Figure 4), scattered endoneurial inflammatory cells, and onion-bulb formation (Figure 5). Treatment with corticosteroids and IVIG resulted in a marked improvement in sensory ataxia in these patients, and they were no longer confined to wheelchairs and they returned to ambulation. Because of the inflammatory hypertrophic demyelinating pathology of CISP and because of the impressive improvement with immunotherapy, CISP is considered a variety of CIDP. Another sensory form of CIDP with more widespread involvement of sensory nerve fibers than CISP (not just sensory roots) has been described as

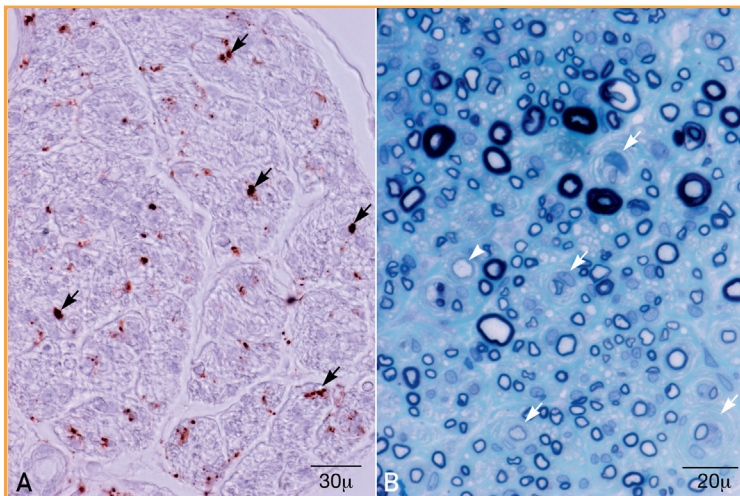


FIGURE 4. Dorsal lumbar rootlet biopsy sections taken from a patient with a restricted sensory form of chronic inflammatory demyelinating polyradiculoneuropathy confined to the sensory nerve root (chronic immune sensory polyradiculopathy). A, Immunostain (CD68) of a paraffin cross section of the dorsal lumbar rootlet showing frequent macrophages within the endoneurium (arrows). Endoneurial macrophage infiltration is abnormal as it is within the blood-nerve barrier. B, Methylene blue preparation of epoxy sections of dorsal lumbar nerve rootlets showing that the density of myelinated fibers is preserved but the size distribution is altered with loss of large myelinated fibers and a relative increase in the number of small myelinated fibers. Note the frequent onion bulbs (arrows) as well as a demyelinated axon surrounded by an onion bulb (arrowhead). The loss of large myelinated sensory fibers shown here correlates well with the clinical syndrome of gait ataxia.

sensory CIPD. Oh et al²³ reported on cases with pure sensory neuropathy on examination, whose electrophysiological testing revealed evidence of demyelination on both motor and sensory studies. All had a progressive course with numbness of their feet and hands, with many developing sensory ataxia. These patients have been reported to respond favorably to immunotherapy.

Although different types of CIPD may be quite focal or distal predominant or proximal predominant, leading to some variability in classification, there are some types of inflammatory demyelinating neuropathies that should be considered separately from CIPD because of likely different underlying pathophysiologies. One inflammatory sensory neuropathy that probably should be categorized apart from CIPD is the distal sensory-predominant neuropathy associated with IgM monoclonal proteins. It is usually clinically mild, but it can cause sensory ataxia. This

IgM-associated neuropathy has been described in association with myelin-associated glycoprotein (MAG) antibodies in more than half the cases. The term *distal, acquired, demyelinating, and symmetrical* (sensory predominant) (DADS) has been used to describe the clinical features of this IgM-associated neuropathy. This distal and sensory predominant phenotype (DADS) is a useful clinical pattern for physicians to recognize as associated with an IgM monoclonal protein, but DADS is not a diagnosis in itself, as it can occur in patients with IgM monoclonal protein and/or MAG antibodies (DADS with MAG) and in patients without monoclonal proteins and MAG antibodies (DADS without MAG). This distinction is important as patients with DADS presentation without MAG or IgM MGUS have a better prognosis than do patients with DADS with MAG or IgM. Except for distal presentation, cases without IgM MGUS act much like typical CIPD and respond nicely to immunotherapy. In contrast, IgM MGUS neuropathy likely has a different underlying pathophysiology than does CIPD and does not respond well to conventional immune-modulating therapy as CIPD does, and so we prefer not to classify it as a form of CIPD, although it is an inflammatory sensory neuropathy that causes distal demyelination and so deserves mention in this discussion on CIPD. Another sensory presentation deserving mention is chronic ataxic neuropathy, ophthalmoplegia, M protein, agglutinins, and disialosyl antibodies syndrome (CANOMAD).²⁴ Patients with CANOMAD present with a sensory neuropathy, ataxia, paralysis of eye movements, and difficulty swallowing and speaking. They have anti-disialosyl antibodies present and are reported to respond favorably to immunotherapy. This syndrome can be thought of as chronic Miller-Fisher syndrome in a similar way that CIPD can be thought of as chronic Guillain-Barré syndrome (AIDP). It is unclear whether CANOMAD cases should be classified as being subtypes of CIPD.

Another area of growing interest in the field of CIPD is the association of this disease process with neurofascin 155 (NF-155) IgG4 antibodies and with contactin 1 antibodies. It is unclear whether neuropathies associated with these paranodal antibodies will ultimately fall under that wide umbrella of CIPD or will

be considered distinct disease processes because of their variability in phenotype and variability in treatment response. Neurofascin 155 IgG4 antibodies are found to be associated with a younger age of onset of sensory ataxia and tremor, and associated central nervous system demyelination has been found in a minority.²⁵ Querol et al²⁶ described these patients with NF-155 antibodies as having a severe phenotype, again commonly with tremor. Patients with CIDP and contactin 1 antibodies have been reported to be older with an aggressive course with early axonal involvement.²⁷

Diagnosis of CIDP

The diagnosis of CIDP is made using a combination of clinical history, physical examination, and electrodiagnostic and laboratory evaluations. Chronic inflammatory demyelinating polyradiculoneuropathy can occur in either a progressive or a relapsing-remitting manner and, as discussed above, is usually a proximal and motor-predominant symmetrical process with accompanying sensory loss. Pain, although reported by some patients, is usually a minor feature. The diagnosis can be more difficult in cases in which the presentation is focal or multifocal or sensory predominant. The diagnosis can be even more challenging later in the disease course when secondary axonal loss can occur, making the characteristic features of demyelination less easy to identify. Chronic inflammatory demyelinating polyradiculoneuropathy can also be misdiagnosed in cases in which there can be mixed demyelinating and axonal changes in electrophysiology secondary to other primary neuropathies, such as diabetic neuropathy. Several criteria have been proposed for the diagnosis of CIDP^{5,17,19,28,29} without an absolute consensus on appropriate criteria. The EFNS/PNS criteria from 2010 are useful and provide guidelines for diagnosis that include a combination of clinical diagnostic criteria (including criteria for atypical CIDP presentations), electrodiagnostic criteria, and supportive criteria (which include CSF and MRI features, sensory electrophysiological findings, nerve biopsy findings, and objective clinical improvement with immunomodulatory therapy).⁶ The accurate diagnosis of CIDP can be challenging, and 1 study of consecutive

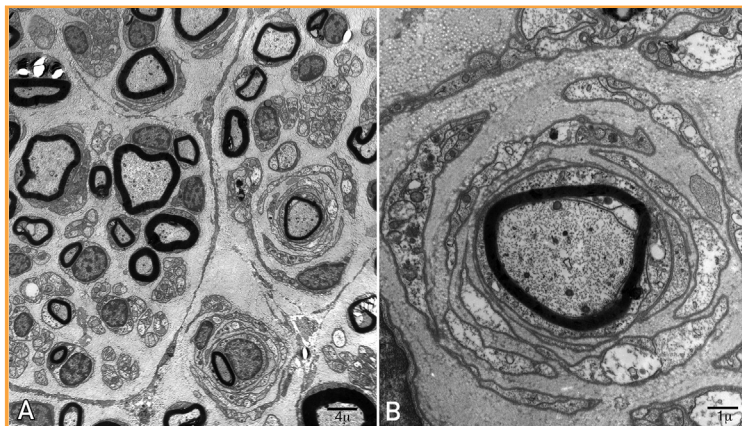


FIGURE 5. Electron micrographs of lumbar dorsal rootlets of a patient with chronic immune sensory polyradiculopathy showing evidence of chronic demyelination and abortive repair. A, Frequent onion-bulb formations (obtained at low power) associated with thinly myelinated and demyelinated profiles. B, Onion bulb at higher power; the stacks of Schwann cell cytoplasmic processes depict the chronicity of this process.

patients referred for a diagnosis of CIDP found that 27 of 58 (47%) did not meet minimal CIDP diagnostic standards when using EFNS/PNS criteria, with the most common of these alternative diagnoses being diabetic peripheral neuropathy (11%), amyotrophic lateral sclerosis (11%), fibromyalgia (11%), and idiopathic small-fiber neuropathy (11%).¹¹

In most cases, laboratory evaluation should include complete blood count, electrolyte levels (including fasting glucose) level, liver function tests, serum and urine monoclonal protein studies including immunofixation, thyroid function tests, angiotensin-converting enzyme, human immunodeficiency virus testing, hepatitis panel, and rheumatological laboratory test results. In some cases (IgM monoclonal proteins present), anti-MAG testing should be considered. If there is a family history, early onset of disease, or lack of response to immunotherapy, genetic testing for an inherited neuropathy (generally Charcot-Marie-Tooth disease Type 1, X-linked Charcot-Marie-Tooth disease, and hereditary neuropathy with pressure palsies) should be considered. Spinal fluid evaluation is also important in evaluation and will typically exhibit cytoalbuminologic dissociation with an elevated CSF protein level and no increase in white blood cell count. A CSF pleocytosis

should raise suspicion for alternative diagnoses such as lymphomatous infiltration, other malignancies, neurosarcoidosis, or infectious etiologies.

In some cases, MRI should be considered to evaluate spinal roots, brachial or lumbosacral plexus, and/or individual nerves. This is usually reserved for atypical cases, often when clinical and electrophysiological findings are focal (such as multifocal CIDP) and are used to rule out other causes of neuropathy and infiltrative pathology. In some cases of CIDP, nerve segments are enlarged because of the hypertrophic nature of the disease with the associated pathological finding of onion bulbs, but this is not a uniform finding. Long-term cases of CIDP are more likely to have this hypertrophic finding. Magnetic resonance imaging of the brain is generally not indicated unless there are associated findings of physical examination or clinical history to suggest superimposed central nervous system dysfunction, as there are reports of cases of coexisting CIDP and multiple sclerosis.³⁰ It should be noted, however, that central nervous system lesions have also been found in some types of inherited neuropathy³¹ and so are not specific for an acquired inflammatory demyelinating etiology. In suspected CIDP cases with an associated monoclonal protein, POEMS syndrome should be excluded with a skeletal bone survey (to look for osteosclerotic myeloma) or positron emission tomography-computed tomography [PET-CT] (to look for osteosclerotic myeloma or Castleman disease) along with vascular endothelial growth factor titers.

Management of CIDP

Management of CIDP is multifaceted, with strategies both targeting the underlying disease process and addressing the functional deficits caused by this disease. Close collaboration with physical medicine physicians is critical to improving patient strength and quality of life, and physical therapy, occupational therapy, and assistive devices for gait (eg, ankle-foot orthoses and walkers) can have a positive effect. The mainstay of treatment, however, is pharmacological interventions. As the primary pathophysiological mechanism of CIDP is inflammatory

demyelination, the pharmacological management has centered on immunosuppressive strategies. The widely accepted first-line management of CIDP consists of 3 therapies: corticosteroids, IVIG, and PLEX. Although these 3 management options can be helpful, each has significant potential adverse effects that can limit their use; and although corticosteroids are generally inexpensive, both IVIG and PLEX are expensive and labor-intensive, lengthy treatments. Multiple other agents are considered second-line therapy and are used primarily when the first-line agents have failed to provide adequate benefit or in an attempt to reduce the need for one of the first-line agents (because of adverse effects, inconvenience to the patient, or cost).

Corticosteroids of various types are frequently used to treat CIDP, and there is a wealth of clinical experience with these agents for the treatment of many immune-mediated diseases and a good understanding of short- and long-term adverse effects. However, evidence-based data on corticosteroid use are limited, although they are still widely considered first-line agents for the initial treatment of CIDP because of the long-term experience in CIDP. The specific type of corticosteroid, type of administration, and dosage varies widely and is largely based on clinical judgment, adverse effect concerns, and accessibility of agents. An early study by Dyck et al³² of 28 patients with CIDP treated with 3 months of oral prednisone reported some improvements in muscle strength and neurological disability compared with no treatment. Another study of patients with CIDP with weekly oral methylprednisolone treatment found remission in 6 of 9 patients after a mean treatment period of 27 months.³³ Another study comparing different immunotherapy regimens found similar improvement rates with the treatments: 81% improvement with intravenous methylprednisolone (IVMP), 88% improvement with oral prednisone or cyclosporine, and 86% improvement with IVIG.³⁴ Boru et al³⁵ retrospectively analyzed 20 patients with CIDP who were treated with 1 g of intravenous methylprednisolone initially daily and then monthly for 5 years and had a 10-year follow-up visit. Of the 15 patients who completed the study, 6 required further

immunotherapy during the 10-year period (in 3, it was IVMP alone with good response). Another study attempted to address whether remission could be achieved by intensive treatments with corticosteroids and assessed patients in a trial comparing 6 monthly doses of dexamethasone (40 mg/d orally for 4 days) with a daily dose of prednisolone (60 mg/d for 5 weeks with tapering to none at the end of 32 weeks). Of this group, cure (defined as >5 years without treatment) or remission (defined as <5 years without treatment) was achieved in 26% after 1 or 2 courses of treatment and occurred in both forms of corticosteroid administration.³⁶ These studies would indicate that corticosteroids in a variety of forms and methods of administration can be effective in the treatment of CIDP but that ongoing treatment is necessary for most patients.

Intravenous immunoglobulin is also frequently used as the initial treatment of CIDP and may be particularly useful in patients with diabetes or patients with other medical comorbidities that would make the use of corticosteroids less ideal. Hughes et al³⁷ performed a large prospective randomized controlled trial evaluating IVIG in 117 patients with CIDP; this included a crossover period for nonresponders and an extension phase for responders. There was a significant improvement in inflammatory neuropathy cause and treatment disability scores after 24 weeks in the IVIG group (54% with IVIG vs 21% with placebo). During the extension phase in the trial, there was a lower rate of relapse in patients treated with IVIG. This study was important in providing strong evidence for the efficacy of IVIG and led to the Food and Drug Administration approval of IVIG for CIDP. It has been debated whether the efficacy of corticosteroids and IVIG is similar. A randomized placebo-controlled study was performed in which patients with CIDP were treated with either IVIG (0.5 g/kg daily for 4 days and then monthly for 6 months) or IVMP (0.5 g daily for 4 days and then monthly for 6 months) and then followed for 6 more months to determine relative tolerability and relapse rate; 52% of patients treated with corticosteroids were able to discontinue treatment, whereas only 13% of patients treated with IVIG were able to discontinue

IVIG—a finding that was statistically significantly different. Eight of the 11 patients who discontinued IVMP did so because of lack of efficacy, whereas only 1 discontinued the medication because of adverse effects. Of the 3 who discontinued IVIG, 2 discontinued because of worsening and 1 because of lack of improvement. There was no significant difference in terms of adverse events. Of the 10 patients who responded to IVMP, none relapsed during the 6 months after cessation of treatment whereas 8 of 21 IVIG responders worsened during the 6 months after cessation of IVIG treatment (a median time of 4 months).³⁸ Another study compared frequency of relapse and length of time to relapse in patients treated with either IVIG or IVMP. They found that over a median follow-up of 42 months, 86% of patients treated with IVIG worsened at a median of 4.5 months after cessation of treatment and 77% of patients treated with IVMP worsened at a median of 14 months after cessation of treatment. Although the percentage of patients relapsing was similar, the difference in time to relapse was statistically significantly more prolonged for corticosteroids.³⁹ Data from these 2 studies suggest that patients with CIDP are more easily weaned off treatment with corticosteroids than with IVIG. However, that conclusion must be made with caution, as there is variability in dosage (amount and frequency) that makes direct comparison difficult in clinical trials, and from a practical standpoint, most clinicians make decisions as to which agent to initiate on a case-by-case basis. However, there is good evidence for the efficacy of both steroids and IVIG.

Subcutaneous immunoglobulins (SCIGs) have been evaluated as an alternative to IVIG. One study took 17 responders to IVIG and transitioned them to SCIG treatment at the same weekly dosage and evaluated them on multiple primary end points including a composite score of muscle performance and function tests (including Medical Research Council score, grip strength, 40-m walking test score, and 9-hole peg test score) and found the overall composite score to be unchanged over the intervals of reevaluation (up to 12 months).⁴⁰ Another trial treated 20 patients with CIDP who had not previously received immunotherapy with either SCIG (0.4 g/kg weekly) for 5 weeks or

IVIg (0.4 g/kg daily) for 5 days, and after 10 weeks, they were crossed over to the other treatment arm. The primary end point was combined isokinetic muscle strength, and there was improvement in both the SCIG and IVIG groups without a significant difference between them. It was noted that the improvement peaked earlier in the IVIG group (at 2 weeks vs 5 weeks in the SCIG group), but the IVIG group was initially given a bolus of IG whereas the SCIG group was not. Therefore, the difference in outcomes seen likely reflects differences in dosage schedule rather than differences in efficacy between SCIG and IVIG. The adverse effects reported with SCIG were found in 3 patients with skin reactions at infusion sites and 2 with nausea.⁴¹ These preliminary results of SCIG are promising for CIDP, but further study is needed using the same dosage schedule. A controlled trial performed in 172 patients with CIDP (PATH study) comparing a low dose (0.2 g/kg) of SCIG, high dose (0.4 g/kg) of SCIG, and placebo was published in 2017.⁴² The study included only patients whose CIDP relapsed when they were weaned off IVIG. They were then again treated with IVIG during a stabilization period and randomized to 1 of 3 treatment arms. During the treatment period, 63% of patients treated with placebo, 39% of patients treated with a low dose of SCIG, and 33% of patients treated with a high dose of SCIG withdrew or relapsed. The absolute risk reduction was 25% for low dose of SCIG vs placebo ($P=.007$) and 30% for high dose of SCIG vs placebo ($P=.001$).

Plasma exchange is the last of the 3 first-line treatments of CIDP. In 1986, Dyck et al⁴³ conducted a double-blinded prospective treatment trial with PLEX vs sham exchanges over a 3-week period in 29 people, resulting in significantly improved combined nerve conduction measurements in patients receiving PLEX. Another trial of 18 patients receiving PLEX found significant improvements in 80% of the 15 patients who completed the trial.⁴⁴ Another prospective observer-blinded study comparing PLEX with IVIG over a 6-week period with a washout and crossover to the alternate treatment group found no significant outcome differences between IVIG and PLEX.⁴⁵ Plasma exchange is still a good and important treatment option in CIDP, but it requires a medical center to

administer it and so for most patients it is not the first-line option. Some experts believe that in a rapidly deteriorating patient, PLEX is the best treatment.

Several oral agents have been used for the treatment of CIDP. One is azathioprine, although a randomized controlled trial of 27 patients treated with prednisone alone vs prednisone together with azathioprine did not report significant outcome differences.⁴⁶ It should be noted that the 9-month treatment period and the low dose of azathioprine used in this study may not have been long enough and high enough to exhibit a significant positive effect. There are other smaller reports of azathioprine efficacy, with 4 of 5 patients in 1 study reporting a benefit of azathioprine although concurrent corticosteroid or corticotropin use could have affected the results,⁴⁷ and another series reported on 3 of 4 patients unresponsive to steroids benefitting from azathioprine. Cocito et al⁴⁸ reviewed data from 110 patients with CIDP (defined as probable or definite using EFNS/PNS 2006 criteria) who were nonresponders to conventional therapy (steroids, immunoglobulins, or PLEX). In this group, they noted that 77 patients were subsequently treated with azathioprine, and 27% of these patients improved with azathioprine. Of note, however, the authors point out that there was no statistically significant difference in response between the alternative immunomodulatory agents used (azathioprine, rituximab, cyclophosphamide, mycophenolate mofetil, cyclosporine, methotrexate, and interferon α).

Gorson et al⁴⁹ performed a retrospective study of mycophenolate mofetil treatment (at a mean dose of 2 g/d for an average of 14 months) in 21 patients with immune-mediated demyelinating neuropathies (12 CIDP, 1 osteosclerotic myeloma, and 8 IgM monoclonal protein with 4 anti-MAG antibodies). They did not report significant improvement between baseline and posttreatment median Medical Research Council scores, sensory scores, or Rankin disability scores. However, the authors suggest that the lack of efficacy may have been related to the patients' long disease course, secondary axonal loss, and general refractoriness. Umapathi and Hughes⁵⁰ reported that 2 of 4 patients with treatment-refractory CIDP treated with

mycophenolate mofetil had minimal improvement. Other small uncontrolled series of CIDP report modest improvement with mycophenolate mofetil.

Oral methotrexate vs placebo was studied in 60 patients with CIDP in a randomized double-blind controlled trial for a 40-week period, with a primary end point of reduction in mean weekly IVIG or steroid dose by more than 20%; 52% of patients treated with methotrexate and 44% of patients treated with placebo reached this primary end point, and the difference was not statistically significant.⁵¹ Seven of 10 patients with treatment-resistant CIDP treated with methotrexate had an improvement in strength of at least 2 points on the Medical Research Council sum score.⁵² A study of cyclophosphamide (intravenous, 1 g/m² monthly for up to 6 months) reported complete remission in 11 of 15 treated patients with CIDP.⁵³ Brannagan et al⁵⁴ reported that 4 patients with refractory CIDP treated with a high dose of cyclophosphamide (200 mg/kg divided over 4 days) all had improvement in strength and functional status. Matsuda et al⁵⁵ reported on 7 patients with treatment-resistant CIDP treated with cyclosporine A, all of whom had significant improvements in disability scores and in grip strength. Barnett et al⁵⁶ retrospectively reviewed 19 patients with CIDP treated with cyclosporine and found that in the progressive CIDP group there was a significant decrease in mean disability and in the relapsing CIDP group there was a significant difference in mean annual incidence of relapse; they noted 2 patients had cyclosporine discontinued because of nephrotoxicity.

Rituximab has shown some efficacy. Some of this work was done on IgM MGUS neuropathies and so will not be discussed here. One study of 13 patients with CIDP with poor response to conventional therapy found that 9 patients responded to rituximab (with a median of 2 months until the start of response).⁵⁷ Several additional cases of CIDP have reported benefits of rituximab.⁵⁸⁻⁶¹ Not all reports have shown benefits of rituximab, though. Gorson et al⁶² reported on 2 patients with IVIG-dependent CIDP who were not able to significantly decrease their IVIG need (defined as a 25% reduction at 1 year after therapy) after rituximab use. However, it should be noted that

these studies are mostly retrospective and uncontrolled.

Other less commonly used therapies have shown some evidence of efficacy in severe cases. These include reported benefits using fludarabine (alone or in combination with other agents). A study of 7 patients with severe IVIG-dependent CIDP treated with alemtuzumab found that 2 patients had a prolonged remission, 2 had a partial response, and 3 had no benefit. However, 3 patients developed autoimmune disease, 1 with autoimmune hemolytic anemia requiring splenectomy with postoperative arrest and death.⁶³ Etanercept was shown to have a “significant benefit” in 3 and “possible improvement” in another 3 of 10 patients with refractory CIDP.⁶⁴ However, there are other reports of the development of CIDP with etanercept use for other purposes.^{65,66} Autologous hematopoietic stem cell transplant (AHSCT) was reported to produce a drug-free remission in 8 of 11 patients who were refractory to other treatment⁶⁷; 3 of the 11 patients relapsed and 1 received a repeat AHSCT. The authors reported, however, that 6 of the 12 transplants were associated with early complications, most especially reactivation of viral and bacterial infections (cytomegalovirus, Epstein-Barr virus, *Escherichia coli*, BK virus, coagulase-negative staphylococci, *Klebsiella pneumoniae*, and alpha-hemolytic Streptococci), so AHSCT should be considered only with severe and difficult-to-treat CIDP cases.

Prognosis of CIDP

Overall, the prognosis of CIDP is reasonably good. One study from 1987 reported on 92 patients with CIDP and indicated that after an average of 10 years of follow-up 73% had good recovery and were independent but noted that 7 patients either died or were immobilized by their disease.⁶⁸ A more recent study of 38 patients with CIDP found that at 5 years after onset of treatment, 87% were still ambulatory, although only 26% had attained complete remission (lasting for >2 years after cessation of treatment). Predictors for complete remission in this study were subacute onset, symmetrical symptoms, lack of muscle atrophy, good response to initial corticosteroid treatment, and a distal pattern of demyelination on nerve conduction studies.⁶⁹ In a

long-term follow-up study of 40 personally followed patients with CIDP treated with IVIG, PLEX, and corticosteroids, Dyck et al⁷⁰ found that all 40 patients exhibited an early response to immunotherapy but that 27 of 37 with late outcome data were still requiring ongoing immunotherapy whereas 10 patients were in remission and not receiving treatment. The authors emphasized that there was often an initial dramatic improvement in treatment in patients with CIDP followed by a small further improvement over the course of time. This may be explained by relatively fast remyelination that can occur followed by slower and less-complete axonal regeneration.

Another issue of importance is whether the subtypes described as different forms of CIDP are truly distinct or represent different presentations of what ultimately becomes a uniform disease process. van Dijk et al⁷¹ reviewed 7 patients with CIDP who had sensory symptoms only at first presentation and found that 5 ultimately developed weakness at a mean duration of 3.1 years. Viala et al⁷² studied 23 patients with multifocal CIDP (Lewis-Sumner syndrome) over a median period of 4 years and found that 9 of those patients had progression of neurological deficits to the other limbs. Attarian et al⁷³ reviewed 15 patients with multifocal CIDP who had been followed at their institution for at least 12 months and noted that in 12 of these patients the pattern continued to be multifocal and asymmetrical. There also appears to be some variability in outcome depending on the clinical phenotype—diffuse vs focal or multifocal. One retrospective study of patients with CIDP found that 64% of patients with classical CIDP had a clinical remission 5 years later as compared with only 41% of patients with multifocal CIDP. The same study indicated that although all the patients with classical CIDP responded to one of the standard treatments (defined as corticosteroids, IVIG, or PLEX), 23% of patients with multifocal CIDP were refractory to any of those treatments.⁶⁹

Patients with CIDP and NF-155 IgG4 antibodies have been reported as having poor response to IVIG.^{25,26} Poor IVIG response has also been described for patients with CIDP and contactin 1 antibodies. Querol et al⁷⁴ reported on 2 patients with anti-NF-155 antibodies and 2 with contactin

1 antibodies poorly responsive to other therapies who were treated with rituximab. They noted significant improvement in 1 patient with NF-155 antibodies and in 1 patient with contactin 1 antibodies. Given the small numbers of patients with NF-155 or patients with contactin 1 systemically studied, detailed treatment recommendations cannot be provided at this time.

Most patients with CIDP do require long-term treatment and follow-up. As with all types of immune-mediated disease, benefits of immunosuppressive treatment have to be balanced with the severity of the underlying clinical presentation and the potential for adverse effects, and the need for immunotherapy as well as its intensity may vary over time. It is our goal in treating CIDP to continually endeavor to wean patients off immunotherapy by periodically following their clinical examinations and nerve conduction studies for worsening. If they do not worsen, then the treatments can be weaned and eventually discontinued, and if there is worsening, then treatments must be increased in dosage and frequency. This is an ongoing titration process, and it is a misconception to think that there is one correct dosage of a drug that is best for CIDP. Rather, the dosage is individually titrated to patients' response. Judgment of treatment response should be made primarily with objective end points (mostly based on the neurological examination and nerve conduction studies) rather than on subjective impressions.

CONCLUSION

Chronic inflammatory demyelinating polyradiculoneuropathy is an inflammatory neuropathy that usually presents with weakness or sensory ataxia. It has variable presentations, including symmetrical (classical), multifocal, sensory predominant, and forms isolated to the sensory roots. Although most cases will need ongoing immunotherapy, some cases do not and can be weaned off treatment. Although challenging because ongoing treatment is frequently required, CIDP is a treatable form of neuropathy and major improvements in the quality of life in patients can be made with ongoing immunosuppression.

Abbreviations and Acronyms: **AHSCT** = autologous hematopoietic stem cell transplant; **AIDP** = acute inflammatory demyelinating polyradiculoneuropathy; **CANOMAD** = chronic ataxic neuropathy, ophthalmoplegia, M protein, agglutinins, and disialosyl antibodies syndrome; **CIDP** = chronic inflammatory demyelinating polyradiculoneuropathy; **CISP** = chronic immune sensory polyradiculopathy; **CSF** = cerebrospinal fluid; **DADS** = distal, acquired, demyelinating, and symmetrical; **EFNS/PNS** = European Federation of Neurological Societies/Peripheral Nerve Society; **IVIG** = intravenous immunoglobulin; **IVMP** = intravenous methylprednisolone; **MAG** = myelin-associated glycoprotein; **MGUS** = monoclonal gammopathy of undetermined significance; **MMN** = multifocal motor neuropathy; **MRI** = magnetic resonance imaging; **NF-155** = neurofascin 155; **PLEX** = plasma exchange; **POEMS syndrome** = polyneuropathy, organomegaly, endocrinopathy, M-protein, skin changes; **SCIG** = subcutaneous immunoglobulin

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